A.

C1 S1 C2 S2 C3 S3 C4 S4 C5 S5

spring cooling
summer drying

spring warming
summer wetting

spring cooling

spring warming

spring warming

Standardized climate level
(calibration SD from the Holocene mean)

0.5
0.0
−0.5

Spring temperature (CCA constraint)
Winter temperature
Summer precipitation (CCA constraint)
Winter precipitation

B.

Fire−frequency ratio
(high fire > 1)

1.2
0.8
0.4

MORA

1.2
0.8
0.4

NOCA

10,000 8,000 6,000 4,000 2,000 0

cal BP